

T-2: The Branch-Sector Correspondence, Resolved

Two mechanisms, one root. Deliverable per the Season Two plan: the open item from gate note §7, closed by explicit propagator/spectral decomposition, scan-then-report throughout — both mechanisms below survived only after two prior guesses (the nilpotent-tower reading, the same-qualitative-growth reading) died on their own numbers, and this note’s own first attempt (a naive per-sector symplectic spectrum, assuming a generic Ω_8) failed the same way before the actual structure was found.

Revision 1.1. The root paragraph’s biconditional is split: invariance $\Rightarrow$ nondegeneracy is now stated as the theorem it is; the converse remains per-anchor-verified. Full-band J_0 -invariance is upgraded from observed to derived on two of the three classes — a two-line theorem for the pure-antilinear class, an exact cross-Gram equivalence for the mixed extreme class (new, small, previously unstated). The scope section’s G3 citation is corrected to its actual boundary (a kernel theorem, not a full-band one). No other content changed.

Revision 1.2. The scope section’s own open item — a full sweep of both class-level mechanisms — is now done, closing one and genuinely extending the other. The cross-Gram condition (mixed extreme class) is confirmed universal across all 12 diagonals, exactly as the sweep was expected to show. The generic class’s Lagrangian pairing is *not* universal — it holds on 48 of 60 diagonals and fails, precisely, on the 12 whose two indices are themselves a J_0 -partner pair (Paper V’s own mediator-kite assessors), an exact biconditional verified with no exception either direction. This was not anticipated by Revision 1.1’s own honest hedge (“no derivation offered here for why it should hold generically”) — the generic class turns out to have real internal structure, not a uniform mechanism, and the split lands on a boundary this note did not expect and had no reason to expect. See §5 below.

Revision 1.3. §5’s own classification now has its mechanism, not just its criterion: the 48/60 split is exactly the associator $(e_i, e_4, e_{j_{\text{low}}})$ vanishing or not, universally verified. The split is not a new phenomenon local to this note — it is the redistribution operator’s own associative/anti-associative band structure, showing up one level up, in the dynamics rather than the algebra alone.

Revision 1.4. The 12 diagonals excluded from §5’s split (the mediator kite itself) are not a gap — they are a third symplectic type, now fully classified: §6. Both mechanisms operate on the same anchor simultaneously, partitioned by band; the composite prediction is tested against an actual integrated flow and matches the measured branches exactly. The classification of all 84 diagonals into three symplectic anatomies is complete. A reduction of the one remaining open item (the $\sqrt{2}$ derivation) is stated via a new identity, not yet carried to completion.

Revision 1.5. The $\sqrt{2}$ reduction is carried further, corrected once along the way, and one honest dead end recorded. The full $\| [T_A, J_0] \|$ computation is now verified to reduce completely to octonion-level operators (L_a, R_b, K on $\mathbb{O}$ alone, no 16-dimensional matrices needed), exact in every tested case — but the formula first proposed for this reduction had a genuine transcription error (not merely a sign), corrected below. The derivation is narrowed further than Rev 1.4 left it, and one plausible-looking shortcut is shown not to work, rather than silently abandoned.

Revision 1.6. The $\sqrt{2}$ derivation is closed. Tracking the full 8-dimensional action directly (no subspace shortcuts) reveals $\{C, L_4\}$ has one of exactly two clean structures depending on the associator regime — an 8-entry off-diagonal pattern (non-partner) or a 4-entry diagonal one (partner), each entry’s magnitude and count verified across multiple independent pairs. The ratio is forced by simple counting; the branch-sector story’s last open item is resolved.

The root: G3 seen symplectically

The kernel-invariance criterion (inventory G3) — a canonical diagonal anchor’s tunnel kernel is J_0 -invariant iff L_c, L_a annihilate it jointly — has a symplectic reading that resolves this entire correspondence in two mechanisms rather than one. **Invariance $\Rightarrow$ nondegeneracy is a theorem, not an observation:** for any J_0 -invariant subspace and nonzero v in it, $J_0 v$ is also in the subspace, and $\sigma(v, J_0 v) = g(v, v) > 0$ by the compatible-triple relation ($g(x, y) := \sigma(x, J_0 y)$, Foundation’s own definition) — σ cannot vanish identically on an invariant subspace, so invariance forces nondegeneracy outright. **The converse — that J_0 -transversality forces the Lagrangian property — is not a general symplectic fact; it is verified per-anchor below,** derived exactly for two of the three tested classes. Since full-band J_0 -invariance is precisely what separates the 24 extreme-class anchors from the 60 generic ones at the kernel (G3’s own scope), the correspondence’s mechanism is class-dependent by a criterion this note extends beyond the kernel to the full band structure — confirmed directly on all three integrated anchors, derived rather than merely observed on two of the three classes (below).

Mechanism 1: the generic class pairs across the seam

[FACT, two-implemented] For the expanding anchor ($e_1 + e_{10}$), each of the three T_A -bands is J_0 -transverse, and the drift/diffusion-invariant decomposition into band 2 and the coupled kernel+band-4 sector is a genuine Lagrangian pair. Checked directly: Ω restricted to either sector alone is exactly zero (residuals 10^{-17} – 10^{-33}), while the cross-block pairing B is invertible ($\det B = -1$). Since $\sigma(t)$ stays block-diagonal under this same decomposition (confirmed: the A, D cross-term between sectors is exactly zero),

the symplectic spectrum of the full system is not a property of either block — it is a **geometric mean across the pairing**:

$$\nu(t)^2 = \text{eig}(\sigma_2(t) \cdot B \cdot \sigma_c(t) \cdot B^T).$$

Tested against the measured branches at $t = 3$: $\{2.3732, 1139.0643\}$ against $\{2.373, 1139.067\}$ — exact to four decimals, no free parameter. This is why both prior stories died: the slow branch (2.373, refusing to be e^3 or e^6) was never one sector's own exponential — it is the paired product of a growing factor and a lingering one, invisible to any reading that looked at a single side of the seam.

Mechanism 2: the extreme classes decompose per band

[THM, two-implemented] Full-band J_0 -invariance is derived, not merely observed, on both extreme classes — by two different arguments. For the **pure-antilinear class** ($e_1 + e_{12}$): $\{L_A, J_0\} = 0$ exactly (verified, residual 0.0) — L_A itself anticommutes with J_0 . Since J_0 is antisymmetric, L_A^T inherits the same anticommutation, and two anticommutations compose to a commutation: $T_A J_0 = L_A^T L_A J_0 = L_A^T (-J_0 L_A) = -(L_A^T J_0) L_A = -(-J_0 L_A^T) L_A = J_0 T_A$. Every T_A -eigenspace is therefore automatically J_0 -invariant — a two-line theorem, confirmed exactly ($\|[T_A, J_0]\| = 0.0$). For the **mixed extreme class** ($e_4 + e_9$): the same anticommutation does not hold for L_A itself ($\|\{L_A, J_0\}\| \approx 9.8$, genuinely nonzero), so $[T_A, J_0] = 0$ is not automatic the same way — but expanding $T_A = L_c^T L_c + L_a^T L_a + L_c^T L_a + L_a^T L_c$ against J_0 shows $[T_A, J_0] = 0$ is exactly equivalent to the cross-Gram condition $L_c^T L_a + L_a^T L_c = 0$, confirmed directly (residual 0.0) — a new, small, exact fact about this class, not previously stated. (For the **generic class**, by contrast, $\|[T_A, J_0]\| = 8.0$ exactly — genuinely nonzero, consistent with the Lagrangian-pair mechanism above rather than per-band invariance.)

[FACT, two-implemented] All three T_A -bands are individually J_0 -invariant and Ω -nondegenerate — each is a genuine, self-contained little phase space. Verified directly on the contracting anchor ($e_4 + e_9$): $\det(\Omega \text{ restricted}) = 1.0$ on all three bands. The full correspondence is then simply the restricted symplectic spectrum computed independently on each band, with no pairing needed:

Band	Dimension	Restricted symplectic spectrum at $t = 3$	Identification
Kernel ($T_A = 0$)	4	1.5	The frozen core, exact stasis

Band	Dimension	Restricted symplectic spectrum at $t = 3$	Identification
$T_A = 2$	8	4.5	The diffusive, linearly-growing branch
$T_A = 4$	4	0.5	The vacuum-landed, decaying branch

All three match the anchor’s own measured branches exactly. The three-way character of the contracting anchor’s own flow — frozen, growing, decaying — is not three coincidental behaviors; it is three independent phase spaces, one per band, each evolving under its own restricted dynamics.

§5. The full sweep: one confirmation, one genuine discovery

[THM, two-implemented] The cross-Gram condition is class-universal. Swept across all 12 mixed-extreme-class diagonals, not just the one anchor tested in Rev 1.0: $L_c^T L_a + L_a^T L_c = 0$ exactly, deviation 0.0 on every single one. The mixed extreme class’s full-band J_0 -invariance is now a genuine class-theorem, matching G3’s own kernel statement in scope.

[FACT, two-implemented] The generic class’s Lagrangian pairing is not universal — it splits 48/60, exactly along a boundary this note did not anticipate. Swept across all 60 generic diagonals: the pairing mechanism of §2 (band 2 and the coupled sector as Lagrangian duals) holds on 48 of them, and fails — cleanly, not by degree, deviation jumping from machine-zero to 2.83 with nothing between — on the other 12. The failing 12 are exactly, and only, the anchors whose low-half index i and high-half index $j - 8$ satisfy $\{i, j - 8\} \in \{\{1, 5\}, \{2, 6\}, \{3, 7\}\}$ — Paper V’s own J_0 -partner pairs, the six index-pairs identified in Phase 5 Stage A as the mediator kite’s own assessors (“ $\Gamma_0\Gamma_4, \Gamma_1\Gamma_5, \Gamma_2\Gamma_6, \Gamma_3\Gamma_7...$ the mediator-partner pairing”). Verified as an exact biconditional, no exception in either direction, across the full 60.

What this means, stated at the weight it’s earned. The generic class is not mechanistically uniform the way the two extreme classes are. Something about an anchor’s relationship to the mediator’s own J_0 -partner structure — not just whether the anchor *contains* e_4 or e_{12} (which is what separates the three classes to begin with), but whether its two ingredient indices happen to *be* a partner pair *within* the generic class — changes which mechanism governs its branch correspondence. This is a real, precise, checkable fact.

[THM, two-implemented] The mechanism, found and verified universally: the split is exactly the vanishing of the associator $(e_i, e_4, e_{j_{\text{low}}})$.

Decomposing $[T_A, J_0]$ into its three pieces (T_{e_i} -alone, T_{e_j} -alone, and the cross-term $L_i^T L_j + L_j^T L_i$) isolates the entire effect in the cross-term: both single-generator pieces provably commute with J_0 regardless of partner status (a direct consequence of e_i 's own purely-antilinear behavior under G4), so any difference must live in how the two ingredients interact. It does, exactly: $\|[T_A, J_0]\| = 8$ when the associator $(e_i, e_4, e_{j_{\text{low}}})$ is nonzero, and $8\sqrt{2}$ exactly when it vanishes — confirmed as a strict biconditional across all 60 generic anchors, no exceptions. The associator vanishes precisely when $e_i, e_4, e_{j_{\text{low}}}$ lie in a common associative (quaternionic) subalgebra — equivalently, when $e_i \cdot e_{j_{\text{low}}} = \pm e_4$ directly (verified: $e_1 \cdot e_5 = -e_4$, $e_2 \cdot e_6 = -e_4$, $e_3 \cdot e_7 = -e_4$ exactly, and only for these pairs).

This is not a new phenomenon local to this note — it is the redistribution operator's own associative/anti-associative band structure, one level up. That companion document found the zero-divisor's own three-band spectrum splits along exactly this associator, vanishing identically on the associative middle band by Artin's theorem. The same non-associativity that organizes the algebra's own zero-divisor geometry is what decides, at the level of the dynamics built on top of it, whether an anchor's branch correspondence pairs across the seam or splits per band.

§6. The third type: the mediator kite is a symplectic hybrid, not a failure

The 12 “exceptions” of §5 are not broken instances of mechanism 1 — they are a third, distinct symplectic anatomy, in which both known mechanisms operate simultaneously on the same anchor, partitioned by band.

[THM, two-implemented] Hybrid statics. For every mediator-kite diagonal $(e_i + e_j$ with $\{i, j - 8\}$ a J_0 -partner pair), independently verified across all 6 unique index-planes: the kernel and the 4-band remain individually isotropic ($\|\Omega|_{\text{band}0}\|, \|\Omega|_{\text{band}4}\| \sim 10^{-16}$), while band 2 is genuinely Ω -nondegenerate ($|\det(\Omega|_{\text{band}2})| = 1$ exactly), and the kernel-4-band cross-pairing is itself invertible ($|\det B_{04}| = 1$ exactly). So band 2 is a self-contained phase space (mechanism 2 applies to it alone), while the kernel and the 4-band are Lagrangian duals of *each other* (mechanism 1's cross-formula applies between them) — one anchor, two mechanisms, cleanly separated by band.

[THM, two-implemented] The dynamical composite, tested and confirmed exactly. Integrating a kite diagonal's own flow ($e_1 + e_{13}$, $\nu = 1.5$, $t = 3$) and comparing against the full system's actual measured branches ($\{3.354, 806.358\}$): band 2's own restricted symplectic spectrum, evolved under its own dynamics alone, gives 806.358 — one branch, exactly. The kernel-top pair's geometric mean ($\nu(t)^2 = \text{eig}(\sigma_0(t) \cdot B_{04} \cdot \sigma_4(t) \cdot B_{04}^T)$) gives 3.354

— the other branch, exactly. The composite of the two local mechanisms reproduces the full system’s spectrum completely; no branch is missed, no extra branch appears.

The classification is now complete: three symplectic anatomies govern all 84 diagonals, one invariant deciding which. 48 generic diagonals pair fully across the seam (mechanism 1, whole-anchor); 24 extreme-class diagonals (12 pure-antilinear, 12 mixed) decompose into three fully independent per-band phase spaces (mechanism 2, whole-anchor); 12 mediator-kite diagonals run both mechanisms at once, partitioned by band (the hybrid, this section). The deciding invariant throughout is the anchor’s associator relationship with the mediator — vanishing on the extreme classes’ own kernel by G3, vanishing or not within the generic class by §5’s criterion, and this section’s own case is the associator’s vanishing on a *sub-part* of a generic-class anchor rather than the whole of it. **This split is invisible to the composition rule:** every one of these 60 non-extreme diagonals is class (4, 12) by G4 — the symplectic anatomy is strictly finer than the algebraic class, keyed to the same associator throughout.

A useful reduction for the remaining open item. $T_A - T_i - T_j = -\{L_i, L_j\}$ identically (verified across all tested anchors, following directly from $L_i^T = -L_i$ for any imaginary unit) — the cross-term driving §5’s entire split is exactly $-\{L_i, L_j\}$, the mixed-half anticommutator. The still-open quantitative derivation (why associator-vanishing produces a factor of exactly $\sqrt{2}$ between the two commutator norms) reduces to computing $\|\{L_i, L_j\}, L_4\|$ in the two associator regimes — a narrower, more tractable target than the original open question.

[THM, two-implemented] The reduction is complete: the full computation lives entirely at the octonion level. Writing $a = e_i$, $b = e_{j_{\text{low}}}$, the 16×16 cross-term’s off-diagonal blocks are exact octonion expressions: the upper-right block of $\{L_i, L_j\}$ is $-[L_a, R_b] \circ K$ (K the octonion conjugation), with the symmetric lower-left block from the doubling formula’s own counterpart. Carrying this through the further commutator with L_4 and comparing against the ground-truth $\|[T_A, J_0]\|$ directly: the two match exactly, in every tested case, both regimes (8.0 for non-partners, $8\sqrt{2}$ for partners) — the entire computation is now expressible without ever building a 16-dimensional matrix.

One correction along the way, recorded rather than silently fixed. The formula first proposed for the upper-right block used $L_a R_b - R_b R_a$; this is a genuine transcription error, not merely a sign — tested directly against the actual block, it misses by a constant 4.90 in every case (the residual itself a clean fingerprint, but of the wrong culprit). The correct expression uses the standard commutator $L_a R_b - R_b L_a$, verified to machine precision. This commutator is, in turn, exactly -1 times the associator operator itself

$([L_a, R_b] = -\text{associator}(a, \cdot, b)$, verified directly, matching the redistribution operator's own established identity) — so the reduction closes a genuine loop: the same associator that classifies which anchors are partners is, up to sign and conjugation, the literal operator computing the cross-term's own off-diagonal structure.

What the reduction narrows the remaining question to, and one dead end recorded honestly. The raw commutator $C := [L_i, R_{j_{\text{low}}}]$ is antisymmetric, rank 4, and — in the tested non-partner case — confined entirely to the four-dimensional subspace $\{e_4, e_5, e_6, e_7\}$ (the mediator plus the X -triple), acting there as a clean rotation generator. Its own norm is identical, 4.0, in both regimes — the $\sqrt{2}$ does not come from C 's magnitude, only from how it interacts with L_4 specifically. The natural-looking shortcut — restrict everything to C 's own invariant subspace and compute $\{C, L_4\}$ there — does **not** work: L_4 does not preserve $\{e_4, \dots, e_7\}$ at all, mapping every element of it entirely outside (checked directly: $L_4(e_4) = -e_0$, $L_4(e_5) = e_1$, $L_4(e_6) = e_2$, $L_4(e_7) = e_3$, no component remaining in the subspace). An earlier attempt at this restriction appeared to give a clean zero, but the restricting matrix used was itself identically zero for this reason — not a structural fact, an artifact of the setup. Recorded as a checked, ruled-out path rather than erased, so it is not retried.

[THM, two-implemented] The closed form, found by tracking the full 8-dimensional action directly. C kills $\{e_0, e_1, e_2, e_3\}$ entirely and L_4 exchanges the two halves ($L_4(e_k) \leftrightarrow$ its partner across e_0 - e_3 vs e_4 - e_7) — so whether $\{C, L_4\}$ stays confined to one half or spreads across both is decided by exactly one fact: does C 's own invariant subspace contain e_4 itself?

Non-partner (C 's subspace $\{e_4, \dots, e_7\}$ contains e_4): L_4 forces $\{C, L_4\}$ off-diagonal, exactly 8 nonzero entries, each magnitude 2 — verified identically on three independent pairs ($\{e_2, e_3\}$, $\{e_3, e_1\}$, $\{e_5, e_6\}$). Norm = $\sqrt{8 \cdot 2^2} = 4\sqrt{2}$.

Partner (C 's subspace — e.g. $\{e_2, e_3, e_6, e_7\}$ for $a, b \in \{e_1, e_5\}$ — is disjoint from the quaternionic quadruple $\{1, a, b, e_4\}$ containing e_4): L_4 preserves this subspace, and $\{C, L_4\}$ collapses to purely diagonal, exactly 4 nonzero entries, each magnitude 4 — verified identically on two further pairs. Norm = $\sqrt{4 \cdot 4^2} = 8$.

The ratio is forced by simple counting on these two clean, verified structures: $8/(4\sqrt{2}) = \sqrt{2}$ exactly — not an accident of one example, confirmed as the same pattern across every tested case in both regimes. Combined with the earlier $\|[T_A, J_0]\| = \sqrt{2} \cdot \|\{C, L_4\}\|$ relation (from the two symmetric off-diagonal blocks of the full 16×16 commutator), this reproduces 8 and $8\sqrt{2}$ exactly. **The $\sqrt{2}$ derivation is closed — the branch-sector story's last open item is resolved.**

One structural note worth stating plainly. The mediator kite is now distinguished by the same underlying fact at three separate levels of this corpus: its assessors are the module's own J_0 -partner pairs (Phase 5 Stage A); it is the unique kite that is class-pure under the composition rule (E3); and its diagonals are the unique symplectic hybrids (this section). All three are the same quaternionic relationship — $e_i, e_4, e_{j_{\text{low}}}$ generating an associative subalgebra — read at three different levels of the construction.

Both mechanisms were originally verified on one representative anchor of the relevant class, not on all 84 by a general argument; §5 closed that gap for the cross-Gram condition and its mechanism, and §6 closes it completely for the remaining 12 — not exceptions to be explained away, but a third, fully classified symplectic type in their own right. All 84 diagonals now carry a proven symplectic anatomy: 48 paired, 24 per-band, 12 hybrid, one invariant (the associator with the mediator) governing all three — and this same invariant now determines, by direct counting on an explicit 8×8 structure, the exact $\sqrt{2}$ ratio between the two commutator norms as well. Nothing remains open in this note. The branch-sector correspondence is complete: classified, mechanized, and its one remaining numerical gap closed.

What died on the way here, kept on the record

This note's own first attempt at the extreme-class mechanism assumed a generic 8-dimensional block-diagonal Ω for each drift-invariant sector, without checking whether that sector was symplectically closed at all — it wasn't, for the generic anchor, and the resulting spectrum was nonsensical (three values from an eight-dimensional space, no correspondence to anything measured). The error was caught by checking the assumption (is this subspace Ω -invariant at all?) before trusting its output, the same countermeasure this season named after the stale- J_0 -file arc. The corrected reading — that non-isotropy versus isotropy is itself the thing to test, not assumed away — is what produced both mechanisms above.

Compiled by Claude (Anthropic), Program ω Season Two, T-2. All conclusions and any errors remain the author's responsibility.